Rosenbrock: gradient-identity kappa versus exact Hessian curvature

Same AdamW trajectories as Analysis016 | Start (0, 1) | 2,000 updates | Weight decay 0.1

$$\widehat{\kappa}_t = \eta^{-1}(1 - \rho_t \|g_t\| / \|g_{t-1}\| + \varepsilon_t), \quad \varepsilon_t = 0$$

Identity kappa proxy (LR color) — Largest Hessian eigenvalue — Exact gradient-direction curvature

(a) $\eta = 10^{-2}$ | unsmoothed

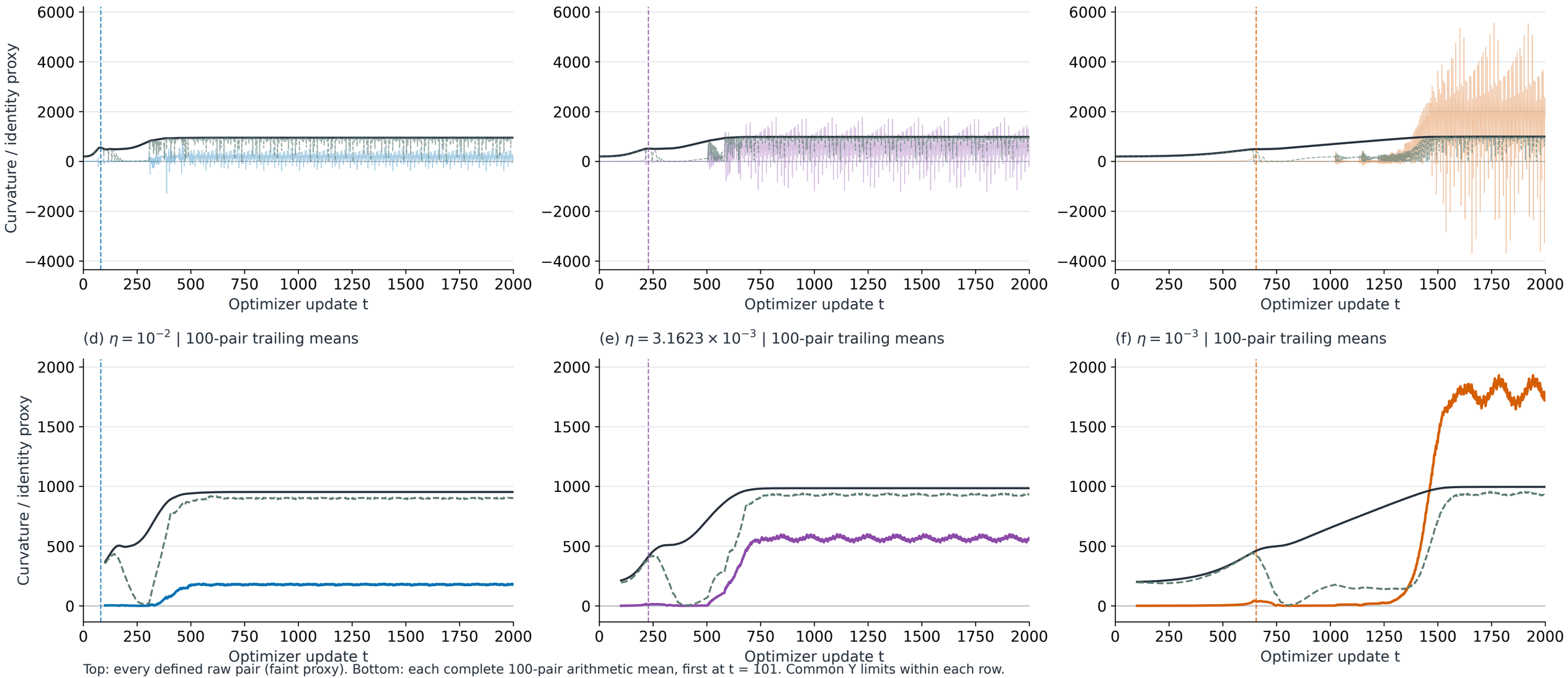
(b) $\eta = 3.1623 \times 10^{-3}$ | unsmoothed

(c) $\eta = 10^{-3}$ | unsmoothed

(d) $\eta = 10^{-2}$ | 100-pair trailing means

(e) $\eta = 3.1623 \times 10^{-3}$ | 100-pair trailing means

(f) $\eta = 10^{-3}$ | 100-pair trailing means



Top: every defined raw pair (faint proxy). Bottom: each complete 100-pair arithmetic mean, first at t = 101. Common Y limits within each row.

For pair t: $g_t = \nabla f(\theta_{t-1})$; Hessian references use $H(\theta_{t-2})$. First pair is undefined. Vertical dashed lines mark original y-turns.

Epsilon is the normalized noise term (zero here), not AdamW's numerical epsilon. No clipping, normalization, thinning or confidence bands.

The identity assumes plain SGD to first order. On AdamW, this is a diagnostic proxy; exact directional curvature is not generally lambda_max.